

Vaccination Data Analysis and Visualization

Completed project report using the supplied vaccination datasets

The project brief requires data cleaning, SQL storage, EDA and Power BI visualization across vaccination coverage, disease incidence, reported cases, vaccine introduction and vaccine schedule data.

Key results

Year	Mean WUENIC coverage
2019	86.93%
2020	84.75%
2021	82.86%
2022	83.50%
2023	84.13%

In the supplied country-level WUENIC observations, the unweighted mean coverage was 84.13% in 2023. This is not a population-weighted global estimate.

Mean DTP1→DTP3 drop-off was 5.07 percentage points in 2023.

2023 MCV1 coverage averaged 85.21% across 195 country observations. The lowest observed values included Montenegro (24%), Democratic People's Republic of Korea (28%), Central African Republic (41%), Yemen (45%) and Somalia (46%).

Coverage/incidence correlations are descriptive and do not establish causality. The source files do not contain gender, education, urban/rural, population density or monthly seasonality fields, so those requested questions are not answered by inference.

Deliverables included in this package

Cleaned CSV data, analysis-ready CSVs, SQL schema and queries, Python pipeline, Power BI dashboard specification, Excel analysis workbook, and this report.